

JWST/NIRSpec IFU 金属梯度候选项目正式汇总

2026-05-29

Executive Summary

分类采用四类:

- A 类: 已有论文利用观测结果得到金属梯度/emission line map;
- B 类: 还没有 published 关注金属梯度/line map 的非 lensed galaxy;
- C 类: 还没有 published 关注金属梯度/line map 的 lensed galaxy;
- D 类: AGN/quasar/radio galaxy candidate, 必须做 AGN subtraction。

1. Metallicity Indicators

indicator	line ratio	strengths	caveats
N2	[NII]6584/H α	单调; 距离近、无需消光改正; scatter 大	受 N/O 比例影响
O3N2	([OIII]5007/H β)/([NII]6584/H α)	同上	低金属端失效
R23	([OII]3727+[OIII]4959,5007)/H β	谱线强	double value degeneracy
R3	[OIII]5007/H β	需要的 line 少, scatter 小	简并问题
O2/R2	[OII]3727/H β	同上	Dust correction; [OII] can be weak.
O32	[OIII]4959,5007/[OII]3727	可以打破简并	对电离也敏感, 不适合作为唯一指标
Ne3O2	[NeIII]3869/[OII]3727	距离近; 单调	对电离也敏感;
S2	[SII]6717,6731/H α	同上	scatter 很大, 只能作为粗略估计
Te/direct	[OIII]4363/[OIII]4959,5007	Least calibration dependent.	[OIII]4363 is faint

2. NIRSpec IFU Wavelength and Redshift Coverage

The ranges below use approximate NIRSpec IFU wavelength coverage and should be read as line-access windows, not sensitivity guarantees.

2.1 Grating/filter wavelength coverage

grating/filter	wavelength range (micron)
PRISM/CLEAR	0.60-5.30
G140/F100LP	0.97-1.89
G235/F170LP	1.66-3.17
G395/F290LP	2.87-5.27

2.2 Single-line redshift coverage

line	rest micron	PRISM/CLEAR	G140/F100LP	G235/F170LP	G395/F290LP
[OII]3727	0.3727	0.61-13.22	1.60-4.07	3.45-7.51	6.70-13.14
[NeIII]3869	0.3869	0.55-12.70	1.51-3.88	3.29-7.19	6.42-12.62
[OIII]4363	0.4363	0.38-11.15	1.22-3.33	2.80-6.27	5.58-11.08
Hbeta4861	0.4861	0.23-9.90	1.00-2.89	2.41-5.52	4.90-9.84
[OIII]5007	0.5007	0.20-9.59	0.94-2.77	2.32-5.33	4.73-9.53
Halpha6563	0.6563	0.00-7.08	0.48-1.88	1.53-3.83	3.37-7.03
[NII]6584	0.6584	0.00-7.05	0.47-1.87	1.52-3.81	3.36-7.00
[SII]6717/31	0.6720	0.00-6.89	0.44-1.81	1.47-3.72	3.27-6.84

2.3 Indicator-set redshift coverage

indicator set	required lines	PRISM/CLEAR	G140/F100LP	G235/F170LP	G395/F290LP	practical note
N2	Halpha+[NII]6584	0.00-7.05	0.48-1.87	1.53-3.81	3.37-7.00	Close line pair; AGN/shock/N/O caveat.
O3N2	Hbeta+[OIII]5007+Halpha	0.61-9.59	1.00-1.87	2.41-3.81	4.90-7.00	Best standard strong-line gradient set for clean HII regions.
R23	[OII]3727+Hbeta+[OIII]5007	0.61-9.59	1.60-2.77	3.45-5.33	6.70-9.53	Double-valued; use O32/R3/O2 priors.
O32	[OIII]5007+[OII]3727	0.61-9.59	1.60-2.77	3.45-5.33	6.70-9.53	Ionization/hardness map, not O/H alone.
Ne3O2	[NeIII]3869+[OII]3727	0.61-12.70	1.60-3.88	3.45-7.19	6.70-12.62	Excitation/hardness diagnostic with auxiliary metallicity information; use with O32, R23 or photoionization modeling.
R3	[OIII]5007+Hbeta	0.23-9.59	1.00-2.77	2.41-5.33	4.90-9.53	High-S/N excitation/oxygen ratio; auxiliary metallicity information.
O2	[OII]3727+Hbeta	0.61-9.90	1.60-2.89	3.45-5.52	6.70-9.84	Low-ionization oxygen ratio; needs dust correction.
S2	Halpha+[SII]6717/31	0.00-6.89	0.48-1.81	1.53-3.72	3.37-6.84	Density/shock/DIG screen; weak at high z.
Te/direct	[OIII]4363+Hbeta+[OIII]5007+[OII]3727	0.61-9.59	1.60-2.77	3.45-5.33	6.70-9.53	Direct-method anchor; [OIII]4363 S/N is limiting.

3. A 类: published metallicity gradient/ line map

(1) Program 3045: Witnessing the Maturing of Teenage Galaxies at $z = 4-6$ with a Comprehensive UV-Optical-Sub-mm Benchmark Sample for the Community

Target.

- Proposal science target(s): DC-494763; DC-519281; DC-536534; DC-630594; DC-683613; DC-709575; DC-742174; DC-842313; DC-848185; DC-873321; DC-873756; VC-5100541407; VC-5100822662; VC-5100994794; VC-5101218326; VC-5101244930; DC-417567; VC-5110377875.
- Physical type: **main-sequence star-forming galaxy sample**.
- Redshift: $4.4 < z < 5.7$.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G235M/F170LP; G395M/F290LP.
- Science Time: ~56.93h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties.:

The ALPINE-CRISTAL-JWST Survey: Gas-phase abundance gradients of main sequence star-forming galaxies and their kinematics at $4 < z < 6$.

- Data/processing: JWST/NIRSpec IFU G235M/G395M (PID 3045,4265,1217,5974), ALMA [CII] 158 μm , NIRCам; IFU reduction, 谱线拟合, 分别利用 Bayesian 方法和线诊断方法得到金属梯度; 使用的 indicators: N2、R3、Ne3O2、O32、O2、S2
- Result: **得到了 20 个 galaxy 的 gas-phase abundance gradients**. 整体梯度接近平坦且 median 略为正, median gradient 为 $+0.039 \pm 0.010 \text{ dex kpc}^{-1}$; 只有 3 个星系在 1σ 水平有 $> 0.05 \text{ dex kpc}^{-1}$ 的正梯度, 没有显著负梯度. 不同 metallicity indicator 得到的梯度没有明显系统偏差.

- Properties: 给出 20 个 galaxies 的物理性质,范围约为 $\log(M_*/M_\odot) \sim 9.19\text{--}10.90$, $\log(\text{SFR}) \sim$

Table 1: Physical properties of the

CRISTAL ID	Full name	$z_{[\text{C II}]}$	R.A. ($^\circ$)	Dec ($^\circ$)
01	DC_842313	4.554	150.2271	2.5
02 [†]	DC_848185, HZ6, LBG-1	5.294	150.0896	2.5
03 [†]	DC_536534, HZ1	5.689	149.9719	2.1
04a	vc_5100822662	4.520	149.7413	2.0
05 [†]	DC_683613, HZ3	5.541	150.0393	2.3
06a	vc_5100541407	4.562	150.2538	1.8
06b	...	4.562	150.2542	1.8
07a [†]	DC_873321, HZ8	5.154	150.0169	2.6
07b	...	5.154	150.0166	2.6
09 [†]	DC_519281	5.575	149.7537	2.0
10a-E [†]	DC_417567, HZ2	5.671	150.5172	1.9
11	DC_630594	4.439	150.1358	2.2
13	vc_5100994794	4.579	150.1715	2.2
14	DC_709575	4.411	149.9461	2.3
15	vc_5101244930	4.580	150.1986	2.3
19	DC_494763	5.233	150.0213	2.0
20	DC_494057, HZ4	5.545	149.6188	2.0
22a	HZ10-E	5.653	150.2471	1.5
25	vc_5101218326	4.573	150.3021	2.3
VC-7875	vc_5110377875	4.551	150.3848	2.4

Notes. ^(a) Derived from SED modellings in Li et al. (2024) and Mitsuhashi et al. (2024). ^(b) about 0.3 dex. ^(c) Classified in Lee et al. (2025a). ^(d) Classified as rotating disk. ^(e) Classified as compact in D. Gómez-Espinoza et al. (in prep.). ^(f) Candidate for detection of H α broad component. ^(g) See Parlanti et al. 2025a for a contrasted signatures of interaction with an extended halo or outflow (Fujimoto et al. 2025a).

1.07—2.75。下面两个 table 分别是性质信息和得到的金属梯度：

ALPINE-CRISTAL-JWST survey: revealing less massive black holes in high-redshift galaxies.

- Data/processing: JWST/NIRSpec IFU(3045)、NIRCam imaging、ALMA [C II] 158 μm ; 在 NIRCam photometric peak aperture 提取 IFU spectrum, 拟合 H α narrow/broad components, 用 broad H α 判断 type-I AGN candidates, 估计 M_{BH} 和 λ_{Edd} 。
- Result: 这篇指出 ALPINE-CRISTAL 样本中有 7 个 type-I AGN candidates, 因此后续做 IFU metallicity gradient 时需要显式处理 broad H α 和 AGN contamination。文章的核心结论是这些 candidates 的 M_{BH} 较低, 整体不像许多高红移 over-massive BH, 而更接近或低于本地 $M_{\text{BH}} - M_*$ 关系。
- Properties: 7 个 type-I AGN candidates; $M_{\text{BH}} \sim 10^6\text{--}10^{7.5} M_\odot$, host M_* 约 $10^{9.5}\text{--}10^{10.5} M_\odot$ 。AGN can-

Table 1. Properties measurements for the AGN candidates and their host galaxies in the ALPINE-CRISTAL-JWST sample

Name	RA	DEC	z^a	H $\alpha^b_{\text{br,corr}}$	FWHM ^c _{Hα,br,corr}	$\log(M_*)^a$	$\log(M_{\text{BH}})^d$	$\log(L_{\text{bol}})$	$\log(L/L_{\text{Edd}})$	E(B-V)
—	deg	deg	—	$10^{-18} \text{ erg s}^{-1} \text{ cm}^{-2}$	km s^{-1}	$M_\odot$	$M_\odot$	erg s^{-1}	—	mag
DC_417567	150.5170078	1.9289283	5.6700	1.69 ± 0.42	596 ± 60	$9.81^{+0.18}_{-0.13}$	6.00 ± 0.10	44.15 ± 0.09	0.05	0.00
DC_519281	149.7537227	2.0910165	5.5759	4.40 ± 0.35	1110 ± 132	$9.87^{+0.13}_{-0.11}$	6.74 ± 0.11	44.49 ± 0.03	−0.35	0.08
DC_536534	149.9718537	2.1181852	5.6886	11.69 ± 1.43	2812 ± 157	$10.36^{+0.11}_{-0.12}$	7.78 ± 0.06	44.88 ± 0.05	−1.01	0.24
DC_683613	150.0392432	2.3372033	5.5420	0.80 ± 0.14	678 ± 112	$10.17^{+0.14}_{-0.12}$	5.95 ± 0.15	43.84 ± 0.07	−0.21	0.00
DC_848185_a	150.0895382	2.5865496	5.2931	1.25 ± 0.20	1618 ± 417	$10.37^{+0.18}_{-0.19}$	6.80 ± 0.23	43.97 ± 0.06	−0.93	0.07
DC_848185_b	150.0896631	2.5863339	5.2931	2.11 ± 0.33	955 ± 206	$10.37^{+0.08}_{-0.19}$	6.43 ± 0.20	44.17 ± 0.06	−0.36	0.20
DC_873321	150.0169147	2.6266375	5.1542	4.27 ± 0.61	986 ± 231	$9.97^{+0.13}_{-0.16}$	6.59 ± 0.21	44.41 ± 0.05	−0.28	0.15

Note.

^a The redshift is measured from far-infrared [CII] line and the M_* is derived from SED fitting (see Sec. 5.1). Both values are taken from Faisst et al. (2020).

^b The broad H α line flux is corrected for the dust extinction using the Balmer decrement.

^c The FWHM of the broad H α line is corrected for the instrumental broadening.

^d The errors for M_{BH} is only the statistical uncertainty propagated from the uncertainties for broad H α line flux and FWHM measurements.

didates 的性质：

Program 5974: ORCHIDS: ORigin of the [C II] Halos In Distant Systems

Target.

- Proposal science target(s): Cristal-05; Cristal-07c; Cristal-09; Cristal-10; Cristal-21.
- Physical type: **LBG**.
- Redshift: $5.1 < z < 5.7$.

Table 3: Metallicity gradients inferred for the CRISTAL samples using multiple strong lines.

ID	Class. ^a	Method I ^b (dex kpc ⁻¹)	Method II ^c (dex kpc ⁻¹)
01a	Non-Disk	0.107 ^{+0.054} _{-0.055}	0.107 ^{+0.028} _{-0.028}
02	Disk	0.031 ^{+0.016} _{-0.017}	0.029 ^{+0.015} _{-0.015}
03	Best Disk	0.065 ^{+0.030} _{-0.031}	0.080 ^{+0.039} _{-0.039}
04a	Non-Disk	0.031 ^{+0.020} _{-0.020}	0.041 ^{+0.058} _{-0.058}
05a	Disk	0.055 ^{+0.058} _{-0.057}	0.049 ^{+0.051} _{-0.051}
06a	Non-Disk	-0.054 ^{+0.083} _{-0.081}	-0.026 ^{+0.054} _{-0.054}
06b	Disk	-0.035 ^{+0.285} _{-0.292}	-0.004 ^{+0.032} _{-0.032}
07a	Disk	0.099 ^{+0.025} _{-0.026}	0.125 ^{+0.027} _{-0.027}
07b	Non-Disk	0.166 ^{+0.091} _{-0.090}	0.079 ^{+0.088} _{-0.088}
09	Disk	0.101 ^{+0.115} _{-0.114}	0.101 ^{+0.099} _{-0.099}
10a-E	Disk	0.017 ^{+0.026} _{-0.027}	0.020 ^{+0.018} _{-0.018}
11	Best Disk	0.025 ^{+0.042} _{-0.040}	0.025 ^{+0.046} _{-0.046}
13	Non-Disk	0.069 ^{+0.037} _{-0.038}	0.029 ^{+0.033} _{-0.033}
14	Non-Disk	0.036 ^{+0.067} _{-0.064}	0.013 ^{+0.025} _{-0.025}
15	Best Disk	0.057 ^{+0.044} _{-0.044}	0.033 ^{+0.061} _{-0.061}
19	Best Disk	0.038 ^{+0.036} _{-0.036}	0.024 ^{+0.045} _{-0.045}
20	Best Disk	-0.029 ^{+0.053^d} _{-0.056}	...
22a	Disk	-0.019 ^{+0.042^d} _{-0.043}	...
25	Non-Disk	0.051 ^{+0.099} _{-0.100}	0.044 ^{+0.029} _{-0.029}
VC-7875	Disk	0.040 ^{+0.024} _{-0.023}	0.039 ^{+0.029} _{-0.029}

Notes. ^(a) Same as Table 1 ^(b) Best-fit metallicity gradients derived from the Bayesian fitting that combines the different diagnostics listed in Table 2. ^(c) Median of the best-fit metallicity gradients derived from each individual diagnostic. The individual values of the gradients are listed in Table F.1. ^(d) Inferred from the metallicity maps in Parlanti et al. (2025a) and Jones et al. (2025).

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G395H/F290LP; G395M/F290LP.
- Science Time: ~43.18h
- Available indicators: N2; O3N2; R3; S2.

Published results and properties.

The ALPINE-CRISTAL-JWST Survey: JWST/IFU Optical Observations for 18 Main-Sequence Galaxies at $z = 4 - 6$. 和 PID 3045 的一样

(2) Program 1893: Galaxy Assembly at $z > 6$: Unraveling the Origin of the Spatial Offset between the UV and FIR Emission

Target.

- Proposal science target(s): BDF3299; COSMOS24108; A2744-YD4.
- Physical type: **star-forming**、多成分的高红移星系系统，其中至少部分位于原星系团中。
- Redshift: 7.114; 6.361; 7.879.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: PRISM/CLEAR.
- Science Time: ~21.3h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties.

Gas-phase metallicity gradients in galaxies at $z \sim 6-8$.

- Data/processing: JWST/NIRSpec IFU, NIRCам/ALMA ancillary data; 拟合发射线, radial/bin-based metallicity-gradient fitting; indicator: R2、O3、Ne3O2、S2、R23
- Result: **得到了二维 emission line map 和 radial metallicity gradient**。主要结论是这些早期系统的 gradients 整体接近平坦, 说明 merger、outflow 或 supernova-driven mixing 可能已经显著抹平早期化学结构。
- Properties: 样本的子结构覆盖 $\log(M_*/M_\odot) \sim 7.6 - 9.3$ 、 $\text{SFR} \sim 1 - 15 M_\odot \text{ yr}^{-1}$ 、 $12 + \log(\text{O}/\text{H}) \sim 7.7 -$

Table 3. Properties of the individual sources.

Source (1)	z (2)	$\log(M_*/M_\odot)$ (3)	$F_{[\text{O II}]}$ (4)	$F_{[\text{Ne III}]}$ (5)	$F_{\text{H}\beta}$ (6)	$F_{[\text{O III}]}$ (7)	$F_{\text{H}\alpha}$ (8)	$F_{[\text{S II}]}$ (9)	SFR (10)
A2744-YD4									
YD4	7.879	$8.69^{+0.18}_{-0.17}$	72 ± 3	<21	36 ± 3	180 ± 4	—	—	$2.2^{+0.2}_{-0.2}$
YD6	7.880	$8.40^{+0.27}_{-0.13}$	26 ± 3	<20	<17	44 ± 3	—	—	<1.0
YD6-[O III]-E	7.881	$7.83^{+0.46}_{-0.22}$	<24	<24	<14	57 ± 3	—	—	<0.87
YD4-[O III]-W	7.877	$7.82^{+0.41}_{-0.48}$	<14	<15	<19	46 ± 3	—	—	<1.1
ZD1	7.872	$7.58^{+0.18}_{-0.37}$	<13	<13	<18	27 ± 3	—	—	<1.1
YD1	7.883	$8.58^{+0.28}_{-0.19}$	107 ± 4	<24	47 ± 3	346 ± 3	—	—	$2.85^{+0.18}_{-0.18}$
YD1-E	7.882	$8.44^{+0.43}_{-0.23}$	54 ± 3	<20	34 ± 2	270 ± 2	—	—	$2.02^{+0.13}_{-0.13}$
s1	7.878	$8.01^{+0.28}_{-0.47}$	<24	<25	18 ± 3	151 ± 3	—	—	$1.09^{+0.16}_{-0.16}$
BDF-3299									
a	7.114	$7.90^{+0.25}_{-0.13}$	24 ± 2	<13	36 ± 1	195 ± 3	—	—	$3.47^{+0.13}_{-0.13}$
b	7.114	$8.21^{+0.30}_{-0.11}$	28 ± 2	<11	44 ± 2	285 ± 4	—	—	$4.2^{+0.2}_{-0.2}$
c	7.112	$7.62^{+0.44}_{-0.14}$	<13	<12	15 ± 2	93 ± 4	—	—	$1.5^{+0.2}_{-0.2}$
COSMOS24108									
a	6.361	$9.29^{+0.09}_{-0.08}$	179 ± 5	48 ± 6	53 ± 5	358 ± 5	219 ± 4	34 ± 4	$14.4^{+1.3}_{-1.3}$
b	6.358	$8.81^{+0.11}_{-0.17}$	150 ± 6	47 ± 6	72 ± 4	470 ± 4	245 ± 3	<18	$9.8^{+0.3}_{-0.3}$
c	6.359	$8.89^{+0.10}_{-0.06}$	67 ± 5	<36	28 ± 4	92 ± 4	68 ± 4	<22	$1.74^{+0.09}_{-0.09}$
[O III]-Ea	6.362	$8.53^{+0.30}_{-0.22}$	62 ± 3	36 ± 3	41 ± 3	296 ± 3	183 ± 3	<21	$14.5^{+1.3}_{-1.3}$
[O III]-Eb	6.356	$8.40^{+0.72}_{-0.32}$	44 ± 4	<24	37 ± 3	244 ± 3	113 ± 3	<22	$3.43^{+0.11}_{-0.11}$

Notes. (1) Source name. (2) Redshift from [O III] $\lambda 5008$. (3) Stellar mass from SED fitting with BAGPIPES. (4-9) [O II] $\lambda 3727, 3$, [O III] $\lambda 5008$, H α , and [S II] $\lambda 6718, 33$ measured fluxes (in units of 10^{-20} erg s $^{-1}$ cm $^{-2}$), (10) SFR (in $M_\odot$ yr $^{-1}$), and (10) gas-phase emission line Gaussian fitting. For the sources in the A2744-YD4 system, the reported M_* and SFR are corrected for the line factor of ~ 2 (Morishita et al. 2023; Bergamini et al. 2023); the line fluxes are instead the observed ones. All the properties are integrated spectra extracted from circular apertures (having radius of 0.15'' for A2744-YD4 and COSMOS24108 and of 0.1'' for BDF-3299) centred on each source. For the line fluxes and SFR, we also report the 3σ upper limits in case of no detection.

8.3. 样本子结构的性质如下:

(3) Program 1840: ALMA [OIII] $\lambda 88\mu\text{m}$ Emitters. Signpost of Early Stellar Buildup and Reionization in the Universe

Target.

- Proposal science target(s): J0235-0532; J0217-0208; SDF-LBG-ID34; RXC-J2248-ID3; COS-2987030247; SXDF-NB1006-2; BDF-3299; A2744-YD4; A1689-ZD1; J1211-0118; COS-3018555981; B14-65666.
- Physical type: [OIII] $\lambda 88\mu\text{m}$ -selected active early galaxy sample.
- Redshift: $6 < z < 7.9$

JWST setup.

- Instrument: NIRSpec/IFU + NIRCам imaging(F115W/F150W/F200W+redshift-dependent LW filters).
- Grating/filter: G395H/F290LP or G395M/F290LP.
- Science Time: ~ 33.9 h
- Available indicators: target-dependent. For the lower-redshift targets, N2; O3N2; S2. For targets at $z \gtrsim 6.7$, R23; R3; O2; Ne3O2; O32. Exact usable sets should be selected target by target.

Published results and properties. 有一些研究金属丰度的论文, 但没有专门计算金属梯度

RIOJA. Young starburst and ionized gas outflows in a $z=7.212$ galaxy uncovered by JWST NIRCам and NIRSpec observations.

- Data/processing: JWST/NIRSpec IFU, NIRCам imaging, ALMA, MIRI; line-map extraction, kinematic/outflow decomposition.
- Result: 得到 **emission line map**. 发现 [OIII] $\lambda 5008$ 有宽线成分, 解释为 ionized-gas outflow system; indicator: O2、R3、R23
- Properties: $\log(M_*/M_\odot) = 8.54^{+0.79}_{-0.22}$; $\log[\text{SFR}/(M_\odot \text{ yr}^{-1})] = 2.54^{+0.17}_{-0.71}$; $12 + \log(\text{O}/\text{H})_{\text{R23}} = 8.07 \pm 0.12$.

(4) Program 1567: Early Galaxy Assembly Uncovered with ALMA and JWST: A Remarkably UV and [CII] Bright, Strongly Lensed Sub-L* Galaxy at z=6.072

Target.

- Proposal science target(s): Group-Z6.3; Z6.1-6.2.
- Physical type: **strongly lensed star-forming galaxy with dense clumps.**
- Redshift: 6.072.

JWST setup.

- Instrument: NIRSpec/IFU + NIRCам imaging(F115W, F150W, F277W, F356W, F444W). Grating/filter: G395H/F290LP.
- Science Time: ~12.3h
- Available indicators: N2; O3N2; R3; S2; Te/direct.

Published results and properties.

Primordial rotating disk composed of at least 15 dense star-forming clumps at cosmic dawn.

- Data/processing: JWST/NIRSpec, NIRCам, ALMA, HST; IFU reduction, 拟合发射线, 得到 line map, sed 拟合, 动力学分析, 分析每个 clump 的性质。indicator: Te direct、R3、N2、O3N2、S2、O3S2。
- Result: **得到了 emission line map**。利用 JWST 图像把一个在 HST image 中看起来较平滑的小星系, 分解成了至少 15 个非常致密的恒星形成团块, 底层结构是一个有序旋转的气体盘

Extended Data Table 3| Intrinsic Physical properties of the Cosmic Grapes on the global

Parameter	Integrated Value	Description
R.A.	06:00:09.5647	Right Ascension (J2000) in the observed frame
Decl.	−20:08:10.993	Declination (J2000) in the observed frame
z_{spec}	6.072	Redshift from [C II] line
μ	$32.5^{+0.7}_{-6.8}$	Average magnification
$r_e^\dagger$ [kpc]	$0.68^{+0.09}_{-0.01}$	Circularized effective radius in rest-frame UV
$M_{\text{UV}}^\dagger$ [mag]	$-19.29^{+0.02}_{-0.26}$	Absolute UV magnitude
$E(B - V)$ [mag]	0.13 ± 0.02	Dust reddening via $H\alpha/H\beta$
n_e [cm ^{−3}]	260^{+400}_{-230}	Electron density via [S II] λ 6718/[S II] λ 6733
SFR [$M_\odot \text{ yr}^{-1}$]	$2.6^{+1.7}_{-1.5}$	Average star-formation rate over 100 Myrs from SED fitting
M_{dyn} [$M_\odot$]	$1.7^{+0.7}_{-0.8} \times 10^9$	Dynamical mass within $2 \times r_e$ from [C II] kinematics
$M_\star$ [$M_\odot$]	$4.5^{+2.7}_{-1.1} \times 10^8$	Stellar mass from SED fitting
M_{dust} [$M_\odot$]	$1.2^{+1.2}_{-0.7} \times 10^6$	Dust mass from FIR SED fitting
M_{gas} ([C II]) [$M_\odot$]	$1.1^{+0.4}_{-0.1} \times 10^9$	Gas mass inferred from [C II] luminosity ¹⁵
M_{gas} (dyn.) [$M_\odot$]	$(1.3 \pm 0.8) \times 10^9$	Gas mass inferred from $M_{\text{dyn}} - M_{\text{star}}$
M_{gas} (dust) [$M_\odot$]	$1.2^{+1.2}_{-0.7} \times 10^9$	Gas mass inferred from the gas-to-dust ($\propto$ metallicity) ¹¹²
T_e ([O III]) [K]	14100 ± 600	Electron temperature via [O III] λ 4363/[O III] λ 5008
12+log(O/H) (T_e)	8.11 ± 0.03	Direct T_e method ^{83, 84}
12+log(O/H) (R3)	8.00 ± 0.09	Strong line method with the R3 index ⁸⁷
12+log(O/H) (N2)	8.30 ± 0.16	Strong line method with the N2 index ⁸⁷
12+log(O/H) (O3N2)	8.21 ± 0.21	Strong line method with the O3N2 index ⁸⁷
12+log(O/H) (S2)	8.03 ± 0.11	Strong line method with the S2 index ⁸⁷
12+log(O/H) (O3S2)	7.98 ± 0.11	Strong line method with the O3S2 index ⁸⁷
log(N/O)	-1.46 ± 0.04	Nitrogen abundance

- Properties: 整体的性质

(5) Program 1554: Nebular Line Diagnostics in a Merger at Cosmic Dawn

Target.

- Proposal science target(s): Group-PJ308-21.
- Physical type: **quasar host and two merging companions.**
- Redshift: 6.234.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G395H/F290LP.
- Science Time: ~7.8h
- Available indicators: N2; O3N2; R3; S2.

Published results and properties.

A quasar-galaxy merger at $z \sim 6.2$: rapid host growth via accretion of two massive satellite galaxies.

- Data/processing: JWST/NIRSpec IFU, ALMA, HST; 扣除了中心 QSO 的 psf, 做了 line map 和 ne 等性质的 map, 但没有量化梯度; indicator: Te direct, N2, R3, S2
- Result: **得到了 emission line map**。把 PJ308-21 解释为 quasar host 正在吸积两个 massive satellites 的 merger system。三个星系的 ISM 差异较大, quasar host 已经很大质量, 但由于 merger 还会快速增长
- Properties: QSO host 的 $\log(M_*/M_\odot) \approx 11$, $Z_{\text{host}} \approx 1.1 Z_\odot$.

(6) Program 1657: Anchoring $z > 6$ Galaxy Metallicities Using Te and ne Diagnostics Enabled by JWST and ALMA Spectroscopy

Target.

- Proposal science target(s): J0218-0519; J0217-0208; J1211-0118.
- Physical type: **$z > 6$ star-forming galaxies, with ALMA [O III] $88 \mu\text{m}$ detection.**
- Redshift: 7.215, 6.204, 6.029.

JWST setup.

- Instrument: NIRSpec/IFU + MIRI/MRS + NIRCам imaging(F150W + F300M; F200W + F410M; F150W + F360M; F200W + F430M). Grating/filter: G235H/F170LP; G395H/F290LP; G395M/F290LP; MIRI MRS Channel 1 SHORT(A).
- Science Time: ~23.7h
- Available indicators: target-dependent. Oxygen-line diagnostics: R23; R3; O2; Ne3O2; O32; Te/direct. N2; O3N2; S2 are available only for targets with $z \lesssim 7.0$.

Published results and properties.

JWST & ALMA Joint Analysis with [OII]3726,3729, [OIII]4363, [OIII]88um, and [OIII]52um: Multi-Zone Evolution of Electron Densities at $z \sim 0 - 14$ and Its Impact on Metallicity Measurements.

- Data/processing: NIRSpec/IFU + NIRCам imaging + ALMA [O III] $88 \mu\text{m}$, [O III] $52 \mu\text{m}$; 发射线拟合, 电子密度计算; indicator: Te direct, R23
- Result: **得到了 line map**。光学线测得的电子密度随红移上升, 但是 FIR 线测得的电子密度却没有明显演化, 说明它们主要追踪不同的气体成分。传统的 Te 方法得到的金属丰度可能被低估, 更偏向 trace 较热、较高密度的气体, 不能代表整个星系。

Table 3. S

Name	SFR ($M_{\odot} \text{ yr}^{-1}$)
J0217-0208	$97.6^{+0.}_{-24}$
SXDF-NB1006-2	$8.1^{+1.}_{-0.}$
J1211-0118	102.4^{+6}_{-2}

NOTE—Errors are 1σ . The SFR is averaged over the past 50 Myr, and the weighted age calculated from the SFR history.

- Properties: metallicity 分别为: $Z_{\text{gas}} \approx 0.32 Z_{\odot}, 0.20 Z_{\odot}, 0.66 Z_{\odot}$.

(7) Program 1827: NIRSpec Integral Field Spectroscopy of LyC-Leaking Galaxies

Target.

- Proposal science target(s): LACES94460; LACES104037.
- Physical type: **LyC-leaking galaxy candidates**.
- Redshift: 3.1

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G235M/F170LP.
- Science Time: ~24.1 h
- Available indicators: N2; O3N2; R3; S2. The O32 value discussed in the related paper should be checked against the observing configuration or ancillary data before being attributed to this IFU setup.

Published results and properties.

Confirmation and Refutation of Lyman Continuum Leakers at $z \sim 3$ with JWST NIRSpec/IFU.

- Data/processing: JWST/NIRSpec IFU (1827) + HST imaging; 确认 LyC 信号区域, 拟合发射线, 星族建模; indicator: Te direct, S2, O32
- Result: **得到了 line map**. 确认了 LyC leaker 区域为 LACES-10403, 该区域由一个年龄仅约 5 Myr 的极年轻星族驱动, merger 可能为 LyC photon escape 提供通道

Table 2. Physical properties of LACES104037

Parameter	Value
T_e^a (K)	13050^{+890}_{-941}
n_e^a (cm^{-3})	239^{+160}_{-132}
$E(B-V)^a$	$0.137^{+0.018}_{-0.020}$
$12 + \log(\text{O}/\text{H})^a$	$8.112^{+0.043}_{-0.043}$
$\log(U)^b$	$-2.41^{+0.10}_{-0.10}$
SFR^c ($M_\odot \text{ yr}^{-1}$)	72 ± 36
$\log_{10}(M_\star/M_\odot)^d$	8.83 ± 0.01
$t_{\text{burst,old}}^d$ (Myr)	15.86 ± 0.02
τ_{old}^d (Gyr)	4.67 ± 0.41
$t_{\text{burst,young}}^d$ (Myr)	< 1

• Properties:

(8) Program 2913: Dissecting the Monsters: Resolved IFU Spectroscopy of the Most Massive Quiescent Galaxies at $z > 3$

Target.

- Proposal science target(s): XMM-VID3-1120; XMM-VID3-2457; XMM-VID1-2075.
- Physical type: **massive quiescent galaxy sample**.
- Redshift: 3.4863, 3.4868, 3.4465.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G235M/F170LP.

- Science Time: ~19.4h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties.

MAGAZ3NE: Spatially Resolved Ages and Chemical Abundances of Ultra-Massive Quiescent Galaxies at $z \sim 3.5$ using JWST/NIRSpec IFU.

- Data/processing: JWST/NIRSpec IFU(2913), Keck spectrum, ALMA dust continuum; 由于是 QG, 主要拟合不同半径处的吸收谱; indicator: $[\alpha/\text{Fe}]$
- Result: **得到了 $[\text{Fe}/\text{H}]$ Map 和金属梯度**. 三个星系中心都经历过强烈 starburst, 中心区域均较年轻, 并且 α -enhanced
- Properties: $\log(M_*/M_\odot) > 11.2$,

(9) Program 5293: Galactic Winds in the Early Universe: observing outflows in emission and absorption in a typical $z \sim 6$ galaxy

Target.

- Proposal science target(s): MACS0308-ZD1.
- Physical type: **strongly lensed clumpy $z=6.2$ star-forming galaxy(Cosmic Spear)**.
- Redshift: 6.208

JWST setup.

- Instrument: NIRSpec/IFU + NIRCам imaging(F115W, F150W, F200W, F250M, F300M, F410M). Grating/filter: G395H/F290LP; G235H/F170LP; G140H/F100LP.
- Science Time: ~10.64h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties.

Spatially Resolved Physical Properties of Young Star Clusters and Star-forming Clumps in the Brightest $z > 6$ Galaxy, the Strongly Lensed Cosmic Spear at $z=6.2$.

- Data/processing: JWST/NIRSpec IFU(5293), NIRCам imaging, ALMA, HST; 发射线提取, sed 拟合;
- Result: **得到了 line map**. 这篇的主要贡献是给强透镜系统的放大率、source-plane morphology 或 clump decomposition. IFU 数据具体处理和金属丰度分析 in prep

Table 1. Measured Properties of the Clumps, Star Clusters, and Counter-image

ID	$R_{\text{eff,obs}}$	$R_{\text{eff,int}}$	$M_{*,\text{obs}}$	$M_{*,\text{int}}$	$\Sigma_{*,\text{int}}$	Age	$\log(Z_*/Z_\odot)$	Π	μ_{tot}	μ_{tan}
(1)	(pix)	(pc)	($10^7 M_\odot$)	($10^7 M_\odot$)	($10^2 M_\odot \text{ pc}^{-2}$)	(Myr)	(8)	(9)	(10)	(11)
1	14.5 ± 0.1	243 ± 2	224^{+75}_{-40}	22^{+8}_{-4}	$6.0^{+2.1}_{-1.1}$	49^{+94}_{-37}	$-0.4^{+0.1}_{-0.1}$	$0.9^{+2.1}_{-0.7}$	10	10
2	5.8 ± 0.1	97 ± 2	29^{+11}_{-6}	$2.9^{+1.1}_{-0.6}$	$4.8^{+1.9}_{-1.0}$	53^{+96}_{-42}	$-0.5^{+0.1}_{-0.1}$	$1.5^{+3.4}_{-1.2}$	10	10
3abc	0.5 ± 0.1	5.9 ± 0.1	15^{+5}_{-9}	$1.1^{+0.4}_{-0.6}$	497^{+181}_{-297}	5^{+2}_{-1}	$-0.7^{+0.3}_{-0.8}$	8^{+5}_{-4}	14	14
4ab	0.9 ± 0.1	5.0 ± 0.1	7^{+6}_{-2}	$0.2^{+0.2}_{-0.1}$	143^{+124}_{-54}	5^{+1}_{-1}	$-1.1^{+0.4}_{-0.3}$	4^{+3}_{-2}	29	29
5ab	0.8 ± 0.2	5.4 ± 1.6	20^{+22}_{-9}	$0.8^{+0.9}_{-0.4}$	396^{+1319}_{-266}	9^{+13}_{-3}	$-0.8^{+0.5}_{-0.7}$	13^{+68}_{-9}	25	24
CI	6.0 ± 0.4	423 ± 31	67^{+26}_{-16}	28^{+11}_{-7}	$0.7^{+0.4}_{-0.2}$	47^{+107}_{-38}	$-0.7^{+0.1}_{-0.2}$	$0.6^{+2.0}_{-0.5}$	2.4	1.3

NOTE—The photometry of star clusters 3, 4, and 5 is represented by the stacked photometry of their multiple images. (1) Clump ID (i.e., label). CI refers to the counter image. (2) Observed and PSF-deconvolved effective radius measured using GALFIT with a 2D Gaussian model. $R_{\text{eff,obs}}$ is calculated as half of the FWHM (full width at half maximum) of the best-fit Gaussian model. The modelling is performed on the F150W image, except for CI which is based on the F200W image because it has stronger detection of the diffuse tail component. The images have spatial sampling of $0''.03 \text{ pixel}^{-1}$. (3) Intrinsic effective radius, corrected for lensing tangential magnification (μ_{tan}). (4) Observed stellar mass derived from SED fitting. (5) Intrinsic stellar mass, corrected for lensing total magnification (μ_{tot}). (6) Intrinsic stellar mass surface density, corrected for lensing. (7) Stellar mass-weighted age. (8) Stellar metallicity. (9) Dynamical age defined as $\Pi \equiv \text{age}/t_{\text{cross}}$ with $t_{\text{cross}} = 10\sqrt{R_{\text{eff}}^3/(G, M_*)}$. (10) Total lensing magnification. (11) Tangential lensing magnification.

- Properties:

(10) Program 1970: Zooming into the Monster' s Mouth: Tracing Feedback from Their Hosts to Circumgalactic Medium in $z=3.5$ Radio-loud AGN

Target.

- Proposal science target(s): TNJ0205+2242; TNJ0121+1320; 4C03.24; 4C19.71.
- Physical type: **$z\sim 3.5$ radio-loud AGN with jet.**
- Redshift: $3.50 < z < 3.59$.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G235H/F170LP.
- Science Time: ~ 24.5 h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties. 目前论文主要做的动力学, 没有做 metallicity

JWST discovers an AGN ionization cone but only weak radiatively driven feedback in a powerful $z\sim 3.5$ radio-loud AGN.

- Data/processing: JWST/NIRSpec, ALMA; 主要处理 IFU 数据, 找 companion 并做了动力学建模
- Result: 有 [O III]5007 channel maps, 追踪电离气体, 最后通过动力学建模得到速度弥散。四个 radio AGN 周围都发现了 companions, 可能主要对应 minor mergers, 触发 radio AGN, 并影响 jet
- Properties: $\log(M_{\star}/M_{\odot})$ 在 10.82-11.27

(11) Program 2654: Kpc-scale Dual Supermassive Black Holes and Their Impact on Galaxy Formation at Cosmic Noon

Target.

- Proposal science target(s): SDSS J0841+4825; SDSS J0749+2255.
- Physical type: **kpc-scale dual quasar system.**
- Redshift: 2.95, 2.17.

JWST setup.

- Instrument: NIRSpec/IFU + MIRI/MRS. Grating/filter: G140M/F100LP; G235M/F170LP; MIRI MRS SHORT; MIRI MRS LONG.
- Science Time: ~ 16.4 h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties.

VODKA-JWST: Synchronized growth of two SMBHs in a massive gas disk? A 3.8 kpc separation dual quasar at cosmic noon with NIRSpec IFU.

- Data/processing: NIRSpec IFU(2654); 主要对 SDSS J0749+2255 双 quasar 处理, 双 psf subtraction, 做二维发射线 map
- Result: 得到了 $H\alpha$ 、[SII]、[NII] flux map。首次直接探测到双 quasar 周围的延展电离气体
- Properties: $M_{\star} \sim 10^{11} M_{\odot}$

(12) Program 2959: Resolving early galaxy disks at $z\sim 8$ with NIRSpec-IFS

Target.

- Proposal science target(s): 04590; 06355; 10612.
- Physical type: **weak lensed early galaxy disk candidates.**
- Redshift: 7.66.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G395H/F290LP.
- Science Time: not reported.
- Available indicators: R23; R3; O2; Ne3O2; O32; Te/direct.

Published results and properties.

Exploring Spatially-Resolved Metallicities, Dynamics and Outflows in Low-Mass Galaxies at $z \sim 7.6$.

- Data/processing: JWST/NIRSpec IFU(2959), NIRCам imaging; 发射线拟合, 计算金属丰度, 动力学分析; indicator: O2、R3、Ne3O2、Te direct
- Result: **得到了 line map 和金属梯度**。使用 strong-line 得到的金属梯度 ~ 0 , 使用 Te direct 得到的金属梯度 $\nabla_Z = -0.11 \pm 0.03 \text{ dex kpc}^{-1}$, 负梯度。作者认为可能原因包括: AGN ionization 污染了 strong-line ratios, strong-line calibration 不适用于 $z \sim 7.6$ 、direct- T_e 方法中的 constant-density assumption 过于简单。
- Properties: 对于 06355 和 10612 分别为: $\log(M_*/M_\odot) = 8.72 \pm 0.04, 8.08 \pm 0.04$, $SFR \sim 54, 15 M_\odot \text{ yr}^{-1}$

(13) Program 2566: Characterizing Stellar Mass Assembly and Physical Properties in the Brightest Galaxy in the Redshift>5 Universe

Target.

- Proposal science target(s): J1241+2219.
- Physical type: **highly magnified $z=5$ lensed galaxy**.
- Redshift: 5.045.

JWST setup.

- Instrument: NIRSpec/IFU + NIRCам imaging(F115W, F150W, F200W, F250M, F335M, F460M). Grating/filter: G395M/F290LP; G235M/F170LP.
- Science Time: $\sim 19.4\text{h}$
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties.

JWST & the Waz Arc I: Spatially Resolving the Physical Conditions within a Post-Starburst Galaxy at Redshift 5 with NIRSpec IFS.

- Data/processing: NIRSpec/IFU(2566) + NIRCам imaging; 发射线提取, 电离参数等性质分析; indicator: N2、O2、R23
- Result: **得到了 line map**。它虽然在 UV 波段非常亮, 但当前发射线整体较弱, Balmer absorption 很强, 说明它曾经经历强烈的恒星形成, 之后快速衰退; 局部低金属丰度可能来自新鲜气体流入。

Table 1. Derived Galaxy Properties

Property	Global
RA, Dec (J2000)	190.3743, 22.5
z_{spec}	5.0453 ± 0.0
Magnification μ_{avg}	92^{+37}_{-31}
$\log_{10}(M_*/M_\odot)$	10.0 ± 0.4
$\log_{10}(\text{sSFR}_{\text{Prospector}}(\text{yr}^{-1}))$	$-9.5^{+0.1}_{-0.2}$

- Properties: $Z_{\text{gas}} \approx 0.5 Z_\odot, \log(M_*/M_\odot) = 9.7 \pm 0.3$

4. B 类: Non-lensed galaxy candidates without published metallicity gradients

(14) Program 1626: Achieving a Revolutionary Panchromatic View of Early Galaxy Growth through NIRSpec/IFU Observations of 12 Massive $z > 6.5$ Galaxies with ALMA-derived [CII] Redshifts

Target.

- Proposal science target(s): REBELS-05; REBELS-08; REBELS-12; REBELS-14; REBELS-15; REBELS-25; REBELS-29; REBELS-32; REBELS-34; REBELS-38; REBELS-39.
- Physical type: **UV 明亮、高质量恒星形成星系，都被 ALMA 探测到 [C II] $158 \mu\text{m}$ 发射线。**
- Redshift: $6.5 < z < 7.7$.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: PRISM/CLEAR.
- Science Time: $\sim 14.1\text{h}$
- Available indicators: target-dependent. R23; R3; O2; Ne3O2; O32 are available across the sample. $\text{H}\alpha$ -based diagnostics such as N2 are available only for the lower-redshift subset and should be selected target by target.

Published results and properties.

REBELS-IFU: evidence for metal-rich massive galaxies at $z \sim 6\text{--}8$.

- Data/processing: JWST/NIRSpec IFU(1626), ALMA; 提取光谱，金属丰度计算，scaling relation 拟合 (MZR 和 FMR); indicators: R23、R3、N2、O32、O2、Ne3O2
- Result: **只得到了整体的金属丰度，没有 line map。**显示这些高红移 massive galaxies 已经有接近成熟的 metal enrichment

Table 4. Summary of the properties derived from the NIRSpec/IFU data analysed in this work. Columns show the nebular attenuation (A_V), $[\text{O III}]\lambda 5007$ EW, O32 ratio, ionisation parameter ($\log U$), oxygen abundance ($12 + \log(\text{O}/\text{H})$), stellar mass ($\log(M_*/M_\odot)$, Stefanon et al. in prep) and star formation rates from the observed $\text{H}\beta$ luminosity ($\text{SFR}_{\text{H}\beta}$). For the four sources at $z \gtrsim 7$, the O32 ratios given are upper limits since no attenuation-correction is applied.

Galaxy	$A_{V,\text{neb}}$ (mag)	$\text{EW}_{[\text{O III}]\lambda 5007}$ ($\text{\AA}$)	O32	$\log U$	$12 + \log(\text{O}/\text{H})$	$\log M_*/M_\odot$	$\text{SFR}_{\text{H}\beta}$ ($M_\odot \text{ yr}^{-1}$)
REBELS-05	0.95 ± 0.42	372 ± 44	2.43 ± 0.31	-2.37 ± 0.04	8.51 ± 0.16	$9.42^{+0.10}_{-0.10}$	137 ± 16
REBELS-08	1.21 ± 0.51	372 ± 95	2.35 ± 0.30	-2.39 ± 0.04	8.22 ± 0.22	$9.33^{+0.07}_{-0.06}$	143 ± 21
REBELS-12	-	387 ± 40	< 5.31	< -2.08	8.23 ± 0.13	$9.54^{+0.03}_{-0.04}$	> 26
REBELS-14	-	616 ± 52	< 7.16	< -1.97	7.90 ± 0.12^a	$9.24^{+0.07}_{-0.06}$	> 45
REBELS-15	0.67 ± 0.32	725 ± 92	9.38 ± 1.20	-1.87 ± 0.07	7.78 ± 0.30	$9.31^{+0.02}_{-0.01}$	204 ± 18
REBELS-18	-	195 ± 27	< 2.67	< -2.34	8.50 ± 0.13	$9.71^{+0.06}_{-0.04}$	> 21
REBELS-25	-	328 ± 19	< 3.42	< -2.25	8.62 ± 0.17	$9.30^{+0.12}_{-0.14}$	> 22
REBELS-29	0.34 ± 0.40	243 ± 22	3.26 ± 0.42	-2.27 ± 0.04	8.73 ± 0.15	$9.69^{+0.07}_{-0.05}$	64 ± 7
REBELS-32	0.68 ± 0.35	378 ± 122	3.10 ± 0.40	-2.28 ± 0.03	8.48 ± 0.13	$9.57^{+0.10}_{-0.08}$	104 ± 10
REBELS-34	1.06 ± 0.74	229 ± 53	2.18 ± 0.28	-2.41 ± 0.05	8.33 ± 0.29	$9.45^{+0.03}_{-0.02}$	68 ± 14
REBELS-38	1.14 ± 0.47	276 ± 45	1.88 ± 0.24	-2.47 ± 0.03	8.28 ± 0.18	$9.75^{+0.09}_{-0.06}$	159 ± 21
REBELS-39	0.62 ± 0.40	599 ± 62	5.05 ± 0.64	-2.10 ± 0.05	8.02 ± 0.29	$9.35^{+0.09}_{-0.08}$	167 ± 19

^a For REBELS-14, the upper and lower branch solutions to the R3 calibration are consistent within the uncertainties with the calibration peak. Therefore, we adopt the average of these solutions as the oxygen abundance and the range between them as the uncertainty.

- Properties:

(15) Program 5572: Red Monsters: Kinematics of Two ‘Universe Breaking’, Ultra-Massive Galaxies in the First Gyr

Target.

- Proposal science target(s): S1; S2.
- Physical type: **ultra-massive early red galaxy sample.**
- Redshift: 5.58, 5.18

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G395H/F290LP.
- Science Time: ~16.8h
- Available indicators: R23; R3; O2; Ne3O2; O32.

Published results and properties. 下面的论文用到了这组观测，但没有关注金属丰度

From Grism to IFU: Revising the Redshift and Nature of the Massive Dusty Galaxy S1 with JWST and ALMA.

- Data/processing: JWST/NIRSpec IFU(5572), ALMA; 只做了谱线识别和 sed, 没有做动力学和金属丰度
- Result: S1 之前在 FRESCO grism 中认为红移是 5.58, 得到 IFU 数据后发现其实是 3.244。
- Properties: 更新后的 S1, $M = 3.6 \times 10^{10} M_{\odot}$, $SFR = 72 M_{\odot} \text{ yr}^{-1}$
- Assessment for metallicity gradients: S1 修正为 $z = 3.244$ 后, G395 setup 错过本报告关注的主要 rest-optical lines, 不作为 metallicity-gradient 推荐对象。

(16) Program 5761: Ionized Gas Kinematics of a $z > 4$ Main Sequence Disk Galaxy

Target.

- Proposal science target(s): ALMA-J081740.86+135138.2.
- Physical type: **massive rotating DLA host disk.**
- Redshift: 4.26.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G395H/F290LP; G395M/F290LP.
- Science Time: ~4.98h
- Available indicators: N2; S2.

Published results and properties. 还没有用这组观测的论文，下面这篇是用另一组 IFU 配置得到的结果，文中说这组观测的数据 in prep

GA-NIFS: A smouldering disk galaxy undergoing ordered rotation at $z = 4.26$.

- Data/processing: NIRSpec/IFU(4258), ALMA; line fitting, metallicity, 动力学建模; indicator: N2、S2
- Properties: $Z_{\text{gas}} \sim 0.7 Z_{\odot}$, $\log(M_{\star}/M_{\odot}) = 10.6 \pm 0.2$

(17) Program 6221: H-alpha mapping of a giant, prototypical Lyman-alpha blob at $z=3$

Target.

- Proposal science target(s): LAB1IFU1.
- Physical type: **giant Lyman-alpha blob.**
- Redshift: 3.1.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G235M/F170LP.
- Science Time: ~24.85h
- Available indicators: N2; O3N2; R3; S2.

Published results and properties. 还没有用到观测的论文

- Properties: 总分子气体质量 $(8.7 \pm 2.0) \times 10^{10} M_{\odot}$, 7 个确认的星系

5. C 类: Lensed galaxy

(18) Program 1864: The Formation of a Primeval Hyperstarburst Galaxy at $z \sim 6$

Target.

- Proposal science target(s): SPT0346-52.
- Physical type: **strongly lensed dusty star-forming galaxy.**
- Redshift: 5.6559.

JWST setup.

- Instrument: NIRSpec/IFU + MIRI/MRS + MIRI imaging(F770W, F1280W, F1500W, F1800W, F2100W) + NIRCам imaging(F200W, F356W, F444W). Grating/filter: NIRSpec G395H/F290LP; MIRI MRS SHORT(A), MEDIUM(B).
- Science Time: ~18.9h
- Available indicators: N2; O3N2; R3; S2.

Published results and properties. 目前没有使用该观测数据的论文

- Properties: $SFR \sim 3000 - 4500 M_{\odot} \text{ yr}^{-1}$, $L_{\text{IR}} \sim 3.6 \times 10^{13} L_{\odot}$

(19) Program 3433: Mapping star formation and feedback in clumpy galaxies at redshift ~5

Target.

- Proposal science target(s): MS1358-ARC; MACS0940-ARC; RCS0224-ARC.
- Physical type: **clumpy strongly lensed star-forming galaxy**.
- Redshift: 4.92, 4.03, 4.88

JWST setup.

- Instrument: NIRSpec/IFU + NIRCам imaging. Grating/filter: G395H/F290LP; G235M/F170LP.
- Science Time: ~38.2h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties. 目前没有用这组观测的论文

- Properties: $M \sim 10^6 - 10^9 M_{\odot}$, UV-derived SFR: $0.1 - 1 M_{\odot} \text{ yr}^{-1}$

(20) Program 5119: Resolving galaxy building blocks at high-z: the comprehensive picture of internal physical properties in an ultra-low-mass major merger system at $z=5.2$

Target.

- Proposal science target(s): ELG1+ELG2.
- Physical type: **ultra-low-mass lensed major merger system**.
- Redshift: 5.1.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G235M/F170LP.
- Science Time: ~24.0h
- Available indicators: R23; R3; O2; Ne3O2; O32.

Published results and properties. 没有用这组观测的论文，但是有一个前置论文

JWST catches the assembly of a $z \sim 5$ ultra-low-mass galaxy.

- Data/processing: NIRCам, HST
- Result: 作者认为 ELG1 和 ELG2 更可能是两个正在并合的星系，而不是一个星系内部的两个 clumps，因为 两个组件之间存在 $H\alpha$ -bright bridge 且 rest-frame optical continuum 中可能有 tidal feature;

Table 1. Inferred properties of the four elements of the lensed galaxy pair. All the physical properties are obtained from 0.3" aperture photometry after applying aperture and lensing corrections. The μ values are from our model. The two rightmost columns are results for combined photometry from the two images.

Property	units	ELG1.1	ELG2.1	ELG1.2	ELG2.2	ELG1 (combined)	ELG2 (combined)
μ		14.1 ^{+0.6} _{-0.4}	16.0 ^{+0.5} _{-0.8}	12.6 ^{+0.3} _{-0.4}	13.5 ^{+0.4} _{-0.5}	26.7 ^{+0.9} _{-0.8}	29.5 ^{+0.9} _{-1.3}
$\log(M_*/M_\odot)$		6.88 ^{+0.15} _{-0.05}	6.84 ^{+0.31} _{-0.25}	6.96 ^{+0.36} _{-0.08}	6.95 ^{+0.35} _{-0.40}	6.98 ^{+0.30} _{-0.14}	6.81 ^{+0.41} _{-0.50}
SFR	$M_\odot \text{ yr}^{-1}$	0.23 ^{+0.00} _{-0.04}	0.12 ^{+0.06} _{-0.11}	0.22 ^{+0.05} _{-0.13}	0.16 ^{+0.07} _{-0.16}	0.23 ^{+0.02} _{-0.03}	0.11 ^{+0.10} _{-0.11}
$\log(\text{sSFR}/\text{Gyr}^{-1})$		1.49 ^{+0.15} _{-0.14}	1.27 ^{+0.46} _{-1.55}	1.39 ^{+0.45} _{-0.45}	1.26 ^{+0.50} _{-2.81}	1.39 ^{+0.32} _{-0.21}	1.25 ^{+0.67} _{-3.27}
A_V	mag	0.21 ^{+0.03} _{-0.05}	0.21 ^{+0.47} _{-0.14}	0.21 ^{+0.50} _{-0.09}	0.24 ^{+0.78} _{-0.12}	0.24 ^{+0.05} _{-0.08}	0.27 ^{+0.95} _{-0.14}
Z_{SED}	$Z_\odot$	0.18 ^{+0.02} _{-0.03}	0.15 ^{+0.05} _{-0.07}	0.15 ^{+0.05} _{-0.08}	0.15 ^{+0.12} _{-0.08}	0.18 ^{+0.02} _{-0.08}	0.15 ^{+0.29} _{-0.09}
H α flux	$10^{-19} \text{ erg s}^{-1} \text{ cm}^{-2}$	2.95 $\pm$ 0.40	1.79 $\pm$ 0.35	2.88 $\pm$ 0.49	2.61 $\pm$ 0.46	2.92 $\pm$ 0.44	2.16 $\pm$ 0.40
H β flux	$10^{-20} \text{ erg s}^{-1} \text{ cm}^{-2}$	10.34 $\pm$ 1.40	6.26 $\pm$ 1.24	10.12 $\pm$ 1.7	9.15 $\pm$ 1.6	10.24 $\pm$ 1.56	7.59 $\pm$ 1.41
[OIII] flux	$10^{-19} \text{ erg s}^{-1} \text{ cm}^{-2}$	9.78 $\pm$ 1.08	6.28 $\pm$ 0.93	10.22 $\pm$ 1.4	7.32 $\pm$ 1.29	9.99 $\pm$ 1.22	6.76 $\pm$ 1.10
$\log \text{EW}_{\text{rest}}(\text{H}\alpha)$		3.16 ^{+0.07} _{-0.08}	3.04 ^{+0.08} _{-0.10}	3.11 ^{+0.08} _{-0.09}	3.07 ^{+0.08} _{-0.09}	3.14 ^{+0.07} _{-0.09}	3.05 ^{+0.08} _{-0.10}
$\log \text{EW}_{\text{rest}}([\text{OIII}]+\text{H}\beta)$		3.38 ^{+0.06} _{-0.06}	3.28 ^{+0.07} _{-0.08}	3.37 ^{+0.07} _{-0.08}	3.23 ^{+0.07} _{-0.08}	3.37 ^{+0.06} _{-0.07}	3.25 ^{+0.07} _{-0.08}
R3		7.08 $\pm$ 1.23	7.51 $\pm$ 1.86	7.56 $\pm$ 1.65	5.99 $\pm$ 1.49	7.31 $\pm$ 1.43	6.67 $\pm$ 1.64
Z_{R3}	$Z_\odot$	0.23 ^{+0.00} _{-0.13}	0.23 ^{+0.00} _{-0.14}	0.23 ^{+0.00} _{-0.13}	0.11 ^{+0.13} _{-0.05}	0.23 ^{+0.00} _{-0.13}	0.16 ^{+0.07} _{-0.09}

- Properties:

(21) Program 5883: The most distant Cosmos-Web strong gravitational lens: mass content in the foreground lens and dissecting the background source

Target.

- Proposal science target(s): CWeb-EinsteinRing.
- Physical type: **strongly lensed background galaxy**.
- Redshift: 5.1043.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G235H/F170LP.
- Science Time: ~10.72h
- Available indicators: for the $z = 5.1043$ background source, R23; R3; O2; Ne3O2; O32.

Published results and properties. 还没有用这组观测的论文

- Properties: background source stellar mass $\sim 1.8 \times 10^{10} M_\odot$; IR-based SFR $\sim 60 M_\odot \text{ yr}^{-1}$.

(22) Program 6405: Clumpy Relics: The First Spectroscopic Confirmation of Globular Clusters at $z \sim 3$

Target.

- Proposal science target(s): UNCOVER35602-Clumps; UNCOVER35602-Host.
- Physical type: **lensed massive QG with surrounding clumps, 可能是 cluster candidate**.
- Redshift: 2.53.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: PRISM/CLEAR.
- Science Time: ~20.27h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties. 没有用到该观测论文

- Properties: Stellar mass 约 $10^6 - 10^7 M_\odot$

6. D 类: AGN / quasar / radio galaxy

(23) Program 1712: JWST Beholds the Multiple-merger Assembly of the Most Luminous Quasar

Target.

- Proposal science target(s): W2246-0526.
- Physical type: **hyperluminous obscured quasar / merger**.
- Redshift: 4.6.

JWST setup.

- Instrument: NIRSpec/IFU + MIRI/MRS. Grating/filter: NIRSpec G235H/F170LP + G395H/F290LP; MIRI MRS SHORT(A), MEDIUM(B), LONG(C).
- Science Time: ~23.3h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties. 目前的论文只用了 MIRI/MRS 的, 且没有做金属丰度相关的

JWST observations and a model for the extremely luminous obscured quasar W2246-0526 at z=4.6.

- Data/processing: MIRI/MRS; 主要用 MIRI 的数据帮助做不同 model 的 sed 拟合, 没有做金属丰度的处理
- Result: 最终通过 sed 拟合得到了各种性质。sed 拟合中需要加上 hot polar-dust component 效果才好
- Properties: $SFR = 360 - 2900 M_{\odot} \text{ yr}^{-1}$; $M_{\text{BH}} = (1.3 - 2.3) \times 10^{10} M_{\odot}$, $M_{*} \sim 5 \times 10^{11} M_{\odot}$.

(24) Program 2028: Mapping a Distant Protocluster Anchored by a Luminous Quasar in the Epoch of Reionization

Target.

- Proposal science target(s): J0910Q; J0910m0414_all_objects_new.
- Physical type: **Luminous broad-absorption-line quasar**.
- Redshift: 6.63.

JWST setup.

- Instrument: NIRSpec/IFU + NIRSpec/MOS. Grating/filter: NIRSpec IFU G395H/F290LP; PRISM/CLEAR for quasar-host continuum.
- Science Time: ~16.3h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32; S2.

Published results and properties. 有使用 MOS 的论文, 但没有使用 IFU

- Properties: $M_{\text{BH}} = (3.59 \pm 0.61) \times 10^9 M_{\odot}$, 位于 protocluster, 富气体、尘埃; $\text{FWHM}_{[\text{C II}] 158 \mu\text{m}} \approx 930 \text{ km s}^{-1}$.

(25) Program 2249: Monster in the Early Universe: Unveiling the Nature of a Dust Reddened Quasar Hosting a Ten-Billion Solar Mass Black Hole at z=7.1

Target.

- Proposal science target(s): J0038-0653.
- Physical type: **dust-reddened high-redshift quasar**.
- Redshift information from proposal_text: proposal redshift: 7.1; 7.06.

JWST setup.

- Instrument: NIRSpec/IFU + MIRI imaging(F560W, F770W, F1000W, F1130W, F1280W, F1500W, F1800W, F2100W, F2550W). Grating/filter: G395H/F290LP; G395M/F290LP; PRISM/CLEAR.
- Science Time: ~5.5h
- Available indicators: R23; R3; O2; Ne3O2; O32.

Published results and properties. 目前没有用到这个观测数据的论文, in prep

- Properties: 估计的 $M_{\text{BH}} \gtrsim 10^{10} M_{\odot}$.

(26) Program 2457: Extreme Quasar Feedback at the Peak of the Galaxy Formation Epoch

Target.

- Proposal science target(s): SDSS-J221524.00-005643.8; SDSS-J083200.20+161500.3; SDSS-J083448.48+015921.1; SDSS-J123241.73+091209.3; SDSS-J121704.70+023417.1.
- Physical type: **extremely red quasar outflow sample**.
- Redshift: $2.40 < z < 2.59$.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G235H/F170LP.
- Science Time: ~20.9h
- Available indicators: N2; O3N2; R3; S2.

Published results and properties. 没有关注金属丰度/梯度的, 主要看 quasar host 的连续谱

JWST IFU observations uncover host galaxy continua in extremely red and obscured quasar.

- Data/processing: JWST/NIRSpec IFU(2457, 1335); quasar PSF subtraction, 拟合形态并提取、拟合 host 一维光谱
- Result: 得到 host 的一维光谱、形态参数、性质。看到了 quasar host 的连续谱, 发现 quasar 与 host 中心经常不重合

Table 3. The derived host galaxy properties from pPXF.

Name	$\log(\frac{\nu L_\nu}{L_\odot})$	M-to- L_V Ratio ($M_\odot/L_\odot$)	$\log(\frac{M_\star}{M_\odot})$ ($M_\odot$)
(1)	(2)	(3)	(4)
SDSSJ0834+0159	10.72 ± 0.11	0.77	10.61 ± 0.33
SDSSJ1232+0912	10.75 ± 0.12	0.77	10.64 ± 0.31
SDSSJ2215−0056	10.94 ± 0.11	0.84	10.87 ± 0.32
XID2028	10.70 ± 0.13	1.68	10.92 ± 0.32
SDSSJ1652+1728	11.19 ± 0.08	0.51	10.90 ± 0.32

NOTE—Column 1: Target name. Column 2: Stellar luminosity of the host galaxy ($L_\odot$) in the restframe Johnson V-band ($\lambda = 0.55\mu\text{m}$) filter (corresponding to observed wavelength of $\sim 1.4-2.1\mu\text{m}$ at $z=1.6-2.9$). Column 3: Stellar mass-to-light ratio in the restframe Johnson V-band filter. Column 4: Stellar mass. The 1σ uncertainties account for systematic errors arising from different Sersic models and a fiducial 0.3 dex uncertainty in the mass-to-light ratios.

• Properties:

(27) Program 3079: BEES: Black hole Extended Emission Search

Target.

- Proposal science target(s): J1335+3533; J158-14; J2100-1715; J2229+1457.
- Physical type: **luminous quasar, with small proximity zone.**
- Redshift: $5.90 < z < 6.15$.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G395M/F290LP.
- Science Time: $\sim 19.64\text{h}$
- Available indicators: N2; O3N2; R3; S2.

Published results and properties. 没有专门测量金属丰度/梯度

BEES: Quasar lifetime measurements from extended rest-optical emission line nebulae at $z \sim 6$.

- Data/processing: JWST/NIRSpec IFU(3079, 1218); psf subtraction, 制作 pseudo-narrowband image, 估计 lifetime
- Result: 得到了 $H\alpha$ 、 $[OIII]$ pseudo-narrowband image。发现多数 quasars 当前 UV-bright phase 很短
- Properties: $M_{\text{BH}} \sim 10^9 M_\odot$

(28) Program 3084: First spatially resolved characterization of a radio-driven outflow at $z \sim 6$

Target.

- Proposal science target(s): P352-15.
- Physical type: **radio-loud quasar with jet-driven outflow.**
- Redshift : 5.83.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G395H/F290LP.
- Science Time: ~ 13.97 h
- Available indicators: N2; O3N2; R3; S2.

Published results and properties. 目前没有使用该观测数据的论文

- Properties: $M_{\text{BH}} \sim 10^9 M_{\odot}$

(29) Program 5192: The first multi-scale and multi-phase characterization of black hole feedback at $z > 6$

Target.

- Proposal science target(s): J0923+0402.
- Physical type: **low-ionization broad-absorption-line quasar.**
- Redshift: 6.626.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G395M/F290LP.
- Science Time: ~ 7.1 h
- Available indicators: N2; O3N2; R3; S2.

Published results and properties. 目前没有使用这个观测的论文

- Properties: $\log(M_{\text{BH}}/M_{\odot}) = 9.4$, $\log(L_{\text{bol}}/\text{erg s}^{-1}) = 47.5$, 有延伸达 15 kpc、包裹着整个宿主星系的巨型 [CII] 气体晕

(30) Program 4877: The Host Galaxy, Environment, and Hot Dust Emission of the First Known Extremely-Luminous Obscured AGN at $z > 6$

Target.

- Proposal science target(s): COS-87259.
- Physical type: **obscured hyperluminous radio-loud AGN with dust-obscured starburst.**
- Redshift: 6.853.

JWST setup.

- Instrument: NIRSpec/IFU + MIRI imaging(F560W, F770W, F1000W, F1130W, F1280W, F1500W, F1800W, F2100W, + NIRCам WFSS(F410M). Grating/filter: PRISM/CLEAR.
- Science Time: ~ 7.21 h
- Available indicators: N2; O3N2; R23; R3; O2; Ne3O2; O32.

Published results and properties. 目前没有使用该观测数据的论文

- properties: $M_{*} \sim 10^{10.24} - 10^{11.2} M_{\odot}$, $L_{\text{IR}} \sim 9 \times 10^{12} L_{\odot}$, $SFR \sim 1300 M_{\odot} \text{ yr}^{-1}$

(31) Program 4912: Mapping the multi-phase outflows in $z \sim 6$ luminous quasars

Target.

- Proposal science target(s): J1148+5251; P183+05; J1319+0950; J2310+1855; J2054-0005.
- Physical type: **$z \sim 6$ luminous quasar outflow sample.**
- Redshift: $6.003 < z < 6.419$.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G395H/F290LP.
- Science Time: 15.32h
- Available indicators: N2; O3N2; R3; S2.

Published results and properties. 目前没有使用该观测数据的论文

- Properties: $M_{\text{BH}} \sim 3.0 \times 10^9 M_{\odot}$

(32) Program 6074: The First Measurement of AGN Feedback in Action in the First Billion Years

Target.

- Proposal science target(s): COSMOSVLA-J100104.51+020203.6.
- Physical type: **heavily obscured radio-loud AGN candidate.**
- Redshift: 7.7.

JWST setup.

- Instrument: NIRSpec/IFU. Grating/filter: G395M/F290LP.
- Science Time: 6h
- Available indicators: R23; R3; O2; Ne3O2; O32.

Published results and properties. 目前没有使用该观测数据的论文

- Properties: $\log(M_{\star}/M_{\odot}) = 11.92 \pm 0.50$, $M_{\text{BH}} \gtrsim 6.4 \times 10^8 M_{\odot}$

7. Recommendation

A 类用于建立方法基准; B/C 类用于寻找新的 star-forming galaxy metallicity-gradient science; D 类需要注意 AGN 的影响

7.1 Published benchmarks

program ID	role	why it matters
3045	Standard gas-phase gradient benchmark	20 个 $4 < z < 6$ main-sequence galaxies 已经得到 gas-phase abundance gradients, 适合作为 reduction、line fitting 和 calibration comparison 的方法基准。
1893	Early-universe resolved gradient benchmark	$z \sim 6 - 8$ systems 已有二维 emission-line maps 和 radial metallicity gradients, 适合研究 merger、outflow 和 mixing 对早期化学结构的影响。
2959	High-redshift calibration stress test	$z \sim 7.6$ low-mass galaxies 已有 resolved maps; strong-line gradient 接近 0, 而 direct- T_e 得到 $\nabla_Z = -0.11 \pm 0.03 \text{ dex kpc}^{-1}$, 可检验高红移 strong-line calibration。

7.2 B 类: non-lensed galaxy recommendations

rank	program ID	target	useful indicators	why recommended	main caveat
1	1626	REBELS $z > 6.5$ galaxy sample	R23; R3; O2; Ne3O2; O32; target-dependent N2	非透镜高红移样本, 适合统计性 resolved enrichment 分析, 并可连接已有 integrated metallicity 结果。	PRISM 分辨率有限; H α diagnostics 只能用于较低红移样本。
2	5761	ALMA-J081740.86+135138.2	N2; S2	旋转盘几何结构清楚, 适合径向 profile 和 gradient 验证。	indicator 较少, 主要依赖 N2; 需谨慎处理 N/O 和 calibration scatter。
3	6221	LAB1IFU1	N2; O3N2; R3; S2	适合研究 LAB 环境中的 line map、component contrast 和 abundance structure。	不是普通 disk-gradient 样本, 应优先讨论环境和不同组件之间的差异。

7.3 C 类: lensed galaxy recommendations

rank	program ID	target	useful indicators	why recommended	main caveat
1	3433	MS1358-ARC; MACS0940-ARC; RCS0224-ARC	N2; O3N2; R23; R3; O2; Ne3O2; O32; S2	多个 clumpy strongly lensed arcs, indicator 丰富, 最适合 source-plane clump-scale abundance mapping。	需要 lens model、source-plane reconstruction 和 clump matching。
2	5119	ELG1+ELG2	R23; R3; O2; Ne3O2; O32	适合比较 merger components 和 H α -bright bridge 的 abundance contrast。	不应强行解释为单一径向 gradient。
3	5883	CWeb-EinsteinRing background source at $z = 5.1043$	R23; R3; O2; Ne3O2; O32	高放大背景源适合 source-plane abundance mapping。	只讨论背景源; 需要可靠 lens reconstruction。

7.4 D 类: AGN / feedback recommendations

rank	program ID	target	useful indicators	why recommended	main caveat
1	1712	W2246-0526	N2; O3N2; R23; R3; O2; Ne3O2; O32; S2	Obscured quasar merger, 适合比较 host、companions 和不同 ionization zones。	必须做 AGN subtraction 等处理。
2	2028	J0910Q protocluster system	N2; O3N2; R3; S2	适合研究 NLR、host 和 companions 的 ionization-abundance structure。	Quasar PSF 和 NLR contamination 明显。
3	4877	COS-87259	R23; R3; O2; Ne3O2; O32	PRISM/CLEAR 覆盖广, 适合探索 obscured radio-loud AGN 的 abundance-ionization mapping。	AGN 和 dusty starburst 污染需要联合建模。